

Prajwal Khanal

p.khanal@utwente.nl · ktm.prajwalkhanal@gmail.com · prajwalkhanal.earth
github.com/prajzwal08 · Google Scholar · ITC Faculty, University of Twente, Enschede, the Netherlands

RESEARCH INTERESTS	Soil–vegetation–atmosphere interaction; water and carbon fluxes; land surface modelling and deep learning, and where the two meet.	
EDUCATION	PhD, Geo-Information Science and Earth Observation, University of Twente Feb 2023—... WUNDER. Process-based land surface modelling and deep learning for water and carbon fluxes. M.Sc., Environmental Engineering, Technical University of Munich Nov 2020—Dec 2022 Graduated 1.4/5.0. Thesis: soil moisture controls on vegetation, in <i>Biogeosciences</i> . B.E., Civil Engineering, Institute of Engineering, Tribhuvan University, Nepal Nov 2014—Aug 2018 Ranked 20/192, 77.7%. Thesis: flood inundation mapping, Babai River.	
EMPLOYMENT	Research Intern, Max Planck Institute for Biogeochemistry, Jena Nov 2020—Feb 2022 Near-surface soil moisture vs. terrestrial water storage, controls on global vegetation. Student Research Assistant, Technical University of Munich Nov 2021—Mar 2022 Flood inundation maps for the Ouémé (Benin) and Naryn (Kyrgyzstan) rivers, HEC-RAS 1D. Working Student, Jena-Geos Ingenieurbüro GmbH, Munich May 2021—Dec 2021 Excel/VBA tool for greenhouse-gas accounting: combustion, refrigeration, transport. Civil Engineer (freelance), ECEP, Kathmandu, Nepal Dec 2018—Dec 2022 Water supply business plans; Lumbini smart-city water master plan. Two ministries.	
PUBLICATIONS	<i>Peer-reviewed</i> Moutahir, H., Sulzer, M., Kiese, R., . . . Khanal, P. , et al. (2026). Matching scales of eddy covariance measurements and process-based modelling: assessing spatiotemporal dynamics of carbon and water fluxes in a mixed forest in Southern Germany . <i>Biogeosciences</i> , 23, 1719–1738. Zeng, Y., Alidoost, F., Schilperoort, B., . . . Khanal, P. , et al. (2025). Towards an open soil-plant digital twin based on the STEMMUS-SCOPE model, following open science . <i>Computers & Geosciences</i> , 205, 106013. Khanal, P. , Hoek van Dijke, A. J., Schaffhauser, T., et al. (2024). Relevance of near-surface soil moisture vs. terrestrial water storage for global vegetation functioning . <i>Biogeosciences</i> , 21(6), 1533–1547. Bhattarai, P., Khanal, P. , Tiwari, P., et al. (2019). Flood inundation mapping of Babai basin using HEC-RAS and GIS . <i>Journal of the Institute of Engineering</i> , 15(2), 32–44. <i>Under review</i> Khanal, P. , Zeng, Y., Daoud, M. G., Beckers, J., Su, Z. Investigating the impact of multilayer resistance schemes on simulating land surface temperature and turbulent fluxes: evaluation within the STEMMUS-SCOPE model . Preprint on SSRN. Khanal, P. , et al. Deep learning soil moisture and fluxes with multimodal foundation-model embeddings. <i>Conference contributions</i> Khanal, P. , et al. (2026). Bridging the scale gap: daily 10-metre soil moisture, evapotranspiration and carbon fluxes via multimodal deep learning. <i>Workshop on Machine Learning for Land and Hydrology</i> , ECMWF, Reading, UK. Khanal, P. , Hoek van Dijke, A., Zeng, Y., Orth, R. (2023). Near-surface vs. sub-surface soil moisture impacts on vegetation functioning . <i>EGU General Assembly</i> , Vienna, EGU23-7670. Khanal, P. (2019). Flood inundation mapping: an inference on the outbreak of flood-borne diseases. <i>International Conference on Health Engineering in Disaster</i> .	
AWARDS	Second place, Hackathon on Machine Learning for Earth System Modelling Aug 2026 Quantization in foundation models. CESOC, ECMWF, TRA Modelling, FZ Jülich, Cologne. Deutschlandstipendium, national merit scholarship, TU Munich 2021/22 Merit-based scholarship, Institute of Engineering, Tribhuvan University 2014—2018	
TEACHING	Teaching Assistant, Technical University of Munich 2021—2022 River Basin Modelling; Integrated Water Resources Management; Rapidly Varying Flow. Assistant Lecturer, Cosmos College of Management and Technology, Nepal Nov 2018—Jan 2021 Elective: Python and GIS for water resources.	
SOFTWARE	WUNDER data viewer, ITC, University of Twente 2026 Weather and soil moisture loggers, read by growers at the food forests as drought develops. Source .	
SKILLS	<i>Modelling:</i> STEMMUS-SCOPE, land surface models, HEC-RAS 1D <i>Data:</i> eddy covariance, satellite soil moisture, ICOS <i>Programming:</i> Python, deep learning, GIS, Excel/VBA <i>Languages:</i> Nepali, English, German	